

2.4

Encoding Memory

How do explicit & implicit memories differ?

Explicit memories, (declarative) are facts & experiences we can consciously know & 'declare'

We encode ExMems through conscious effortful processing

But some info skips the conscious encoding track, and goes into storage.

This automatic processing produces our implicit memories (nondeclarative)

Our 2-track mind helps encode, retain & retrieve through both effortful & automatic tracks.

What info do we process automatically?

Im Mem Procedural memory for automatic skills
 Classically Conditioned Associations

Without conscious effort, we auto process:

Space: Ex. While studying, we often encode location in notebook where info is stored

Time - Throughout day, we auto note sequence of events. ex. later, if we realize losing something, this allows us to retrace steps.

Frequency - We keep track of how many times something happens. ex. realizing running into same person twice in 1 day

Our 2nd mind has **very efficient** info-processing.

As one track fucks away routine details

Other track free to **focus on conscious, effortful, processing**

Vision, thinking, mem not single ability: we **split info** into 2 tracks

Skills, like learning to read, is **originally explicit**, but **after practice** became automatic.

What is sensory memory?

Sen Mem records **momentary** images, sounds & strong scents.

Sen Mem is very fleeting.

Iconic Memory, & SenMem, $\therefore$ fleeting senmem of visual stimuli.

For few tenths of a second, our eyes register a picture-image memory of a scene, which we can recall in amazing detail - until it very quickly fades.

Our Echoic Memory, is a fleeting SenMem for auditory stimuli.

EchMem lasts $\sim 3-4$ seconds.

How much memory can we store in short-term memory?

We can recall $\sim 7 \pm 2$ bits of info from STmem

Exact # varies by task (~ 6 letters, ~ 5 words)

Without active processing, STmems have short lives

Young adults have more Wmem capacity.

What are some tricks to aid memory?

Chunking.

Chunking info into familiar segments aids recall

Chunking happens automatically, when we read words - chunks of letters

Mnemonics

Ancient greek scholars & orators developed mnemonics to help encode lengthy passages of speech.

Many of these use vivid imagery, because we are very good at remembering mental pictures.

∴ We more easily remember concrete, visualizable words than we do abstract words

The Peg-word system requires memorizing a jingle associating numbers with words that rhyme with them.

One-bun, two-shoe, etc.

Then you create a mental picture combining each item on a list with an item associated with a number.

The method of Loci is a very powerful technique.

It involves placing items at specific locations in a familiar location.

Hierarchies

When we develop expertise in a field, we process info in hierarchies composed of few broad categories, divided & subdivided into narrower concepts & facts.

Taking notes in class in an outline format is like a hierarchy.

What are optimal ways to learn?

We retain info better when encoding is distributed over time.

Spacing Effect:

Massed practice (cramming) can produce speedy short-term learning & inflated confidence, but we forget this info quickly.

But Distributed practice produces better long-term recall.

This effect extends to motor skills & online game performance.

Testing effect:

repeated self-testing, practicing retrieval, is far more effective than re-reading material.

Levels of processing

We process info at varying levels

depth of processing affects our long term retention.

Shallow Processing encodes on an elementary level

incl. word's letters :: structural encoding
or slightly more intermediate level, a word's sound :: phonemic encoding

Deep Processing encodes semantically, based on meaning of words

more meaningful processing → better retention

Making Material Personal

When info isn't meaningful nor personal, it's harder to recall.

We typically recall not literal text but what we encoded

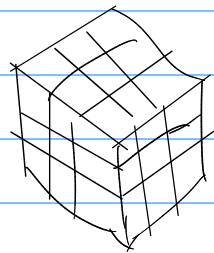
We can avoid big mismatches by rephrasing what we see and hear into meaningful terms

very useful to relate to previous material

Self-reference effect, tendency to remember self-relevant info, is very powerful in hyper-individualistic western cultures.

Collectivist eastern cultures tend to remember self-relevant & family-relevant info equally well.

Personal Connection



I used to solve rubik's cubes blindfolded
(Scrambling, then memorizing, then blind solving)

One technique I found very consistent involved the method of loci. By assigning letters to squares & making words to place around my house.

MCQs:

- | | |
|------|-------|
| 1. a | 2. c. |
| 3. a | 4. b |
| 5. d | 6. d. |